FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF

Experiment (4)
Diode Rectifiers with Peak Detector

Objectives

- To study the bridge rectifier function.
- To analyze the output of a peak detector.

Pre-lab Work

Analyze the circuit below using the constant voltage diode model. The input is a sinusoidal wave with amplitude V_m and frequency f. Find V_o with and without the capacitor C. In each case, draw the waveform of the output and find the average output voltage and average load current.

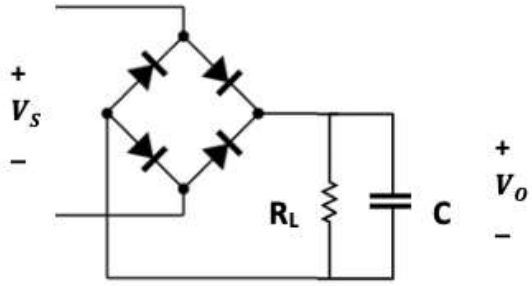


Figure 1: Full-wave bridge rectifier with peak detector.

Solution

Equipment

Table 1. Equipment for experiment 4.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	1 kΩ Resistor (2W, 5%)			
3	Capacitors (3)			
4	Diodes (4)	1N4001		
5	Multi-meter (1)			
6	Dual trace Oscilloscope			
7	Probes and cables			

Experimental Procedure

Part (1): Full-wave Rectifier Output Analysis.

1. Connect the circuit in figure 1 without the capacitor and with R_L=1 kΩ. Pay a special attention to how you connect the terminals of the four diodes. The input voltage source V_S is sinusoidal. Note: V_S should be connected as shown not to ground.

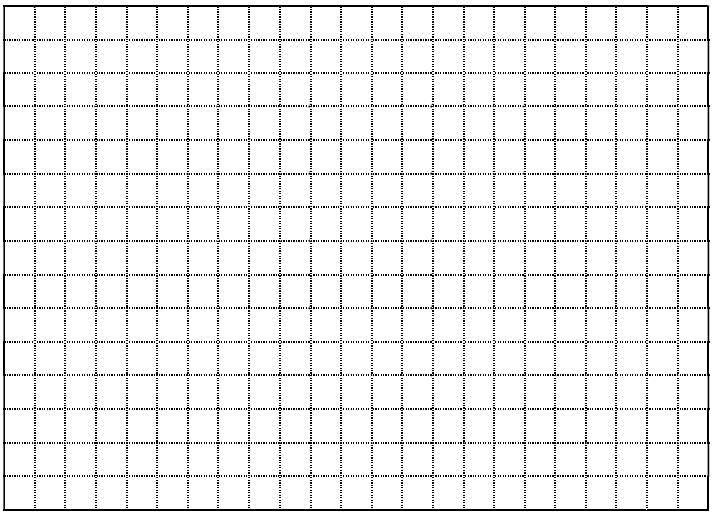
2. Connect channel 1 of the oscilloscope to V_S and adjust it to 10 V amplitude at 60 Hz. Pay attention to the polarity of the oscilloscope connection. Sketch the input waveform in graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible.
3. Now connect channel 2 of the oscilloscope to V_O. Sketch the output waveform on the same graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible. Use different colors for the sketches.
4. Using the multi-meter, measure the output voltage and record it in table 2.
5. Repeat steps 1-4 for a triangular V_S waveform. Use graph 2 for the sketches.

Part (2): Full-wave Rectifier with Peak Detector.

1. Consider again the circuit in figure 1 with the capacitor connected.
2. Repeat the whole procedure in part 1 for both capacitors provided. Record results in graphs 1 and 2 and table 3. Use different colors for the sketches.

Results and Analysis

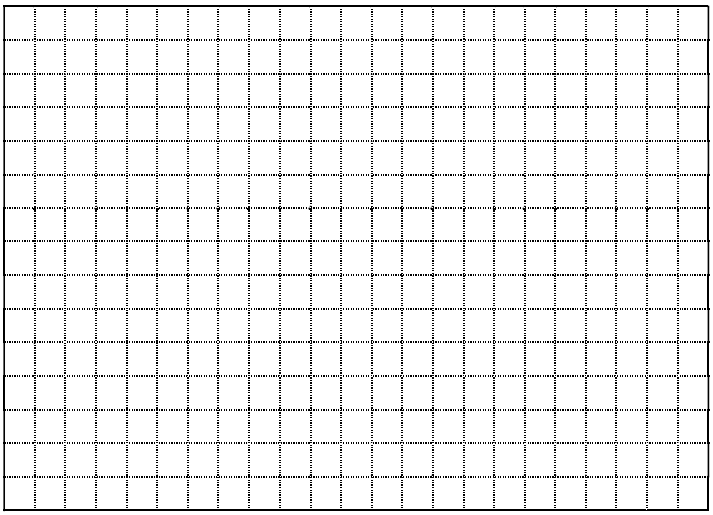
Part (1): Full-wave Rectifier Output Analysis.



Graph 1: Sinusoidal input case.

Table 2. Average output voltage of the bridge rectifier without peak detector.

Sinusoidal input			Triangular input		
$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference



Graph 2: Triangular input case.

1. Calculate the theoretical value of the output voltage for both input waveform types and record it in table 2.
2. Comment on your results regarding the waveforms in graphs 1 and 2.

3. Comment on your results regarding the average output voltage.

Part (2): Full-wave Rectifier with Peak Detector.

Table 3. Average output voltage of the bridge rectifier with peak detector.

	Sinusoidal input			Triangular input		
Capacitor value (μF)	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference

1. Calculate the theoretical value of the output voltage for both capacitor values and record it in table 3.
2. Comment on your results regarding the waveforms in graphs 1 and 2.

3. Comment on your results regarding the average output voltage

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							